National University of Computer and Emerging Sciences, Lahore Campus

	Course:	DLD	Name:	
	Program:	BS(SE)	Semester:	spring-2026
	Duration:	30 minutes	Total Marks:	25
	Paper Date:	10-02-2026	Weight	
	paper:	A	Page(s):	
	Exam:	Quiz 1	Roll No:	

Q1. Number System

a) Perform **hexadecimal multiplication**:
(A3) × (1F)
Show complete working and write the final answer in **hexadecimal**.

b) Using **2's complement**, subtract:
(110101) − (011110)
Also state whether the result is **positive or negative** and why.

Q2. Boolean algebra – Simplification & De Morgan

Simplify the following Boolean expression using **Boolean identities only**:

F = (A + B'C)(A' + B + C)(A + C')

a) Simplify the expression step by step.
b) Apply **De Morgan's laws** on the original expression and write the resulting form.
c) State **which identities** are used during simplification

Q3. Boolean Laws

a) What is the **Consensus Theorem**? Write its general form and simplify:

c) Explain **why the consensus term can be removed**, in your own words.

Q4. SOP / POS

Given the simplified Boolean expression:

F(A, B, C) = A + B'C

a) Convert the given expression into **proper SOP form**.
b) Convert the same expression into **proper POS form**.
c) Clearly mention **which form has more terms** and why.

A·1·1
A(B+B̄)
AB+AB̄

Q5. Truth Table & Circuit Design

A digital circuit has three inputs A, B, C.
The output **F = 1** when **any two inputs are 1s**

a) Construct the **truth table** for the circuit.
b) Write the **Boolean expression** for F.
c) Draw the **logic circuit diagram** using basic gates.